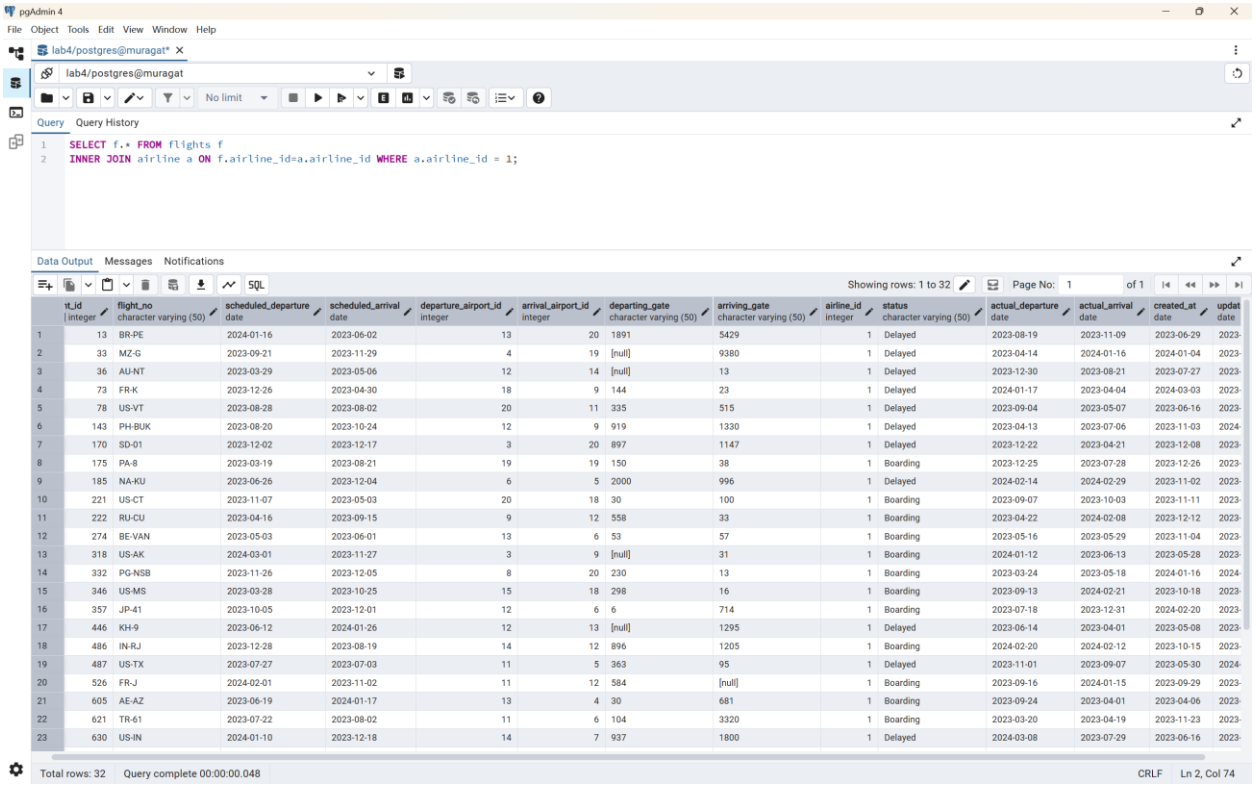
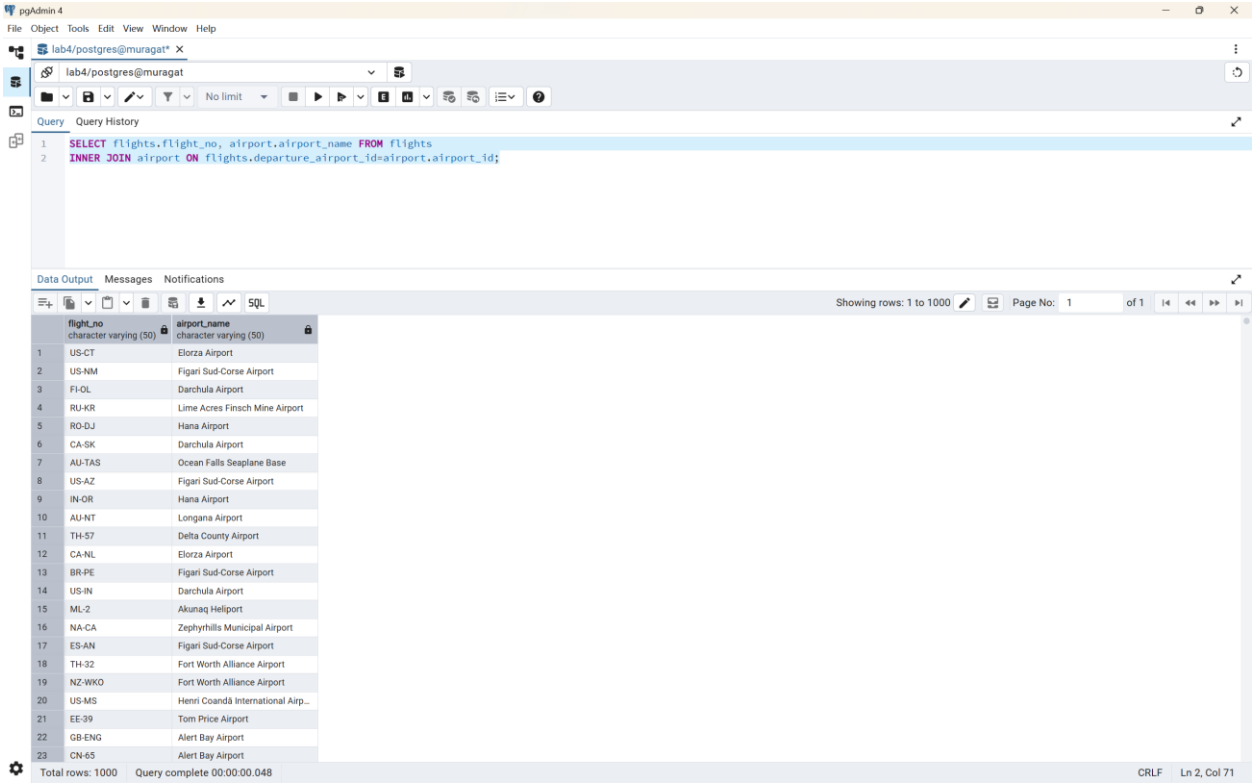
Laboratory work 6

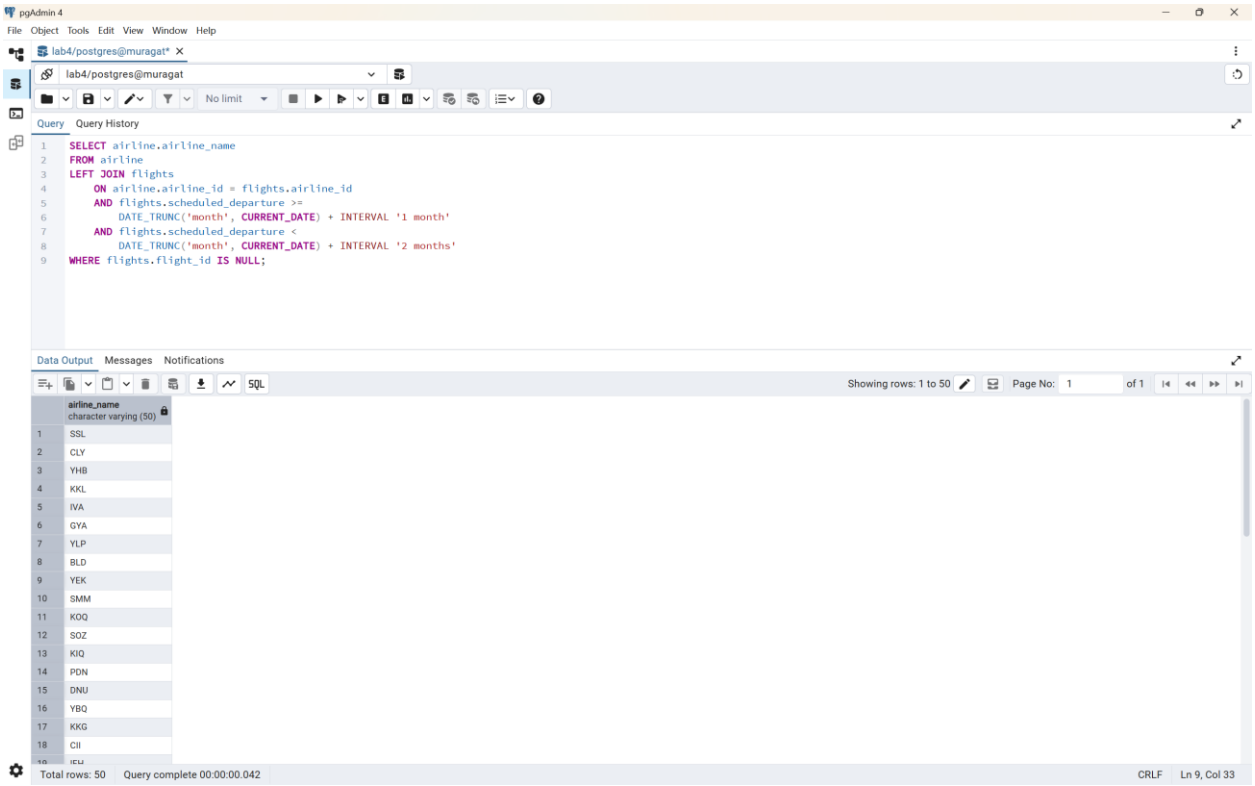
1. Write a query that displays all flights of a specific airline.



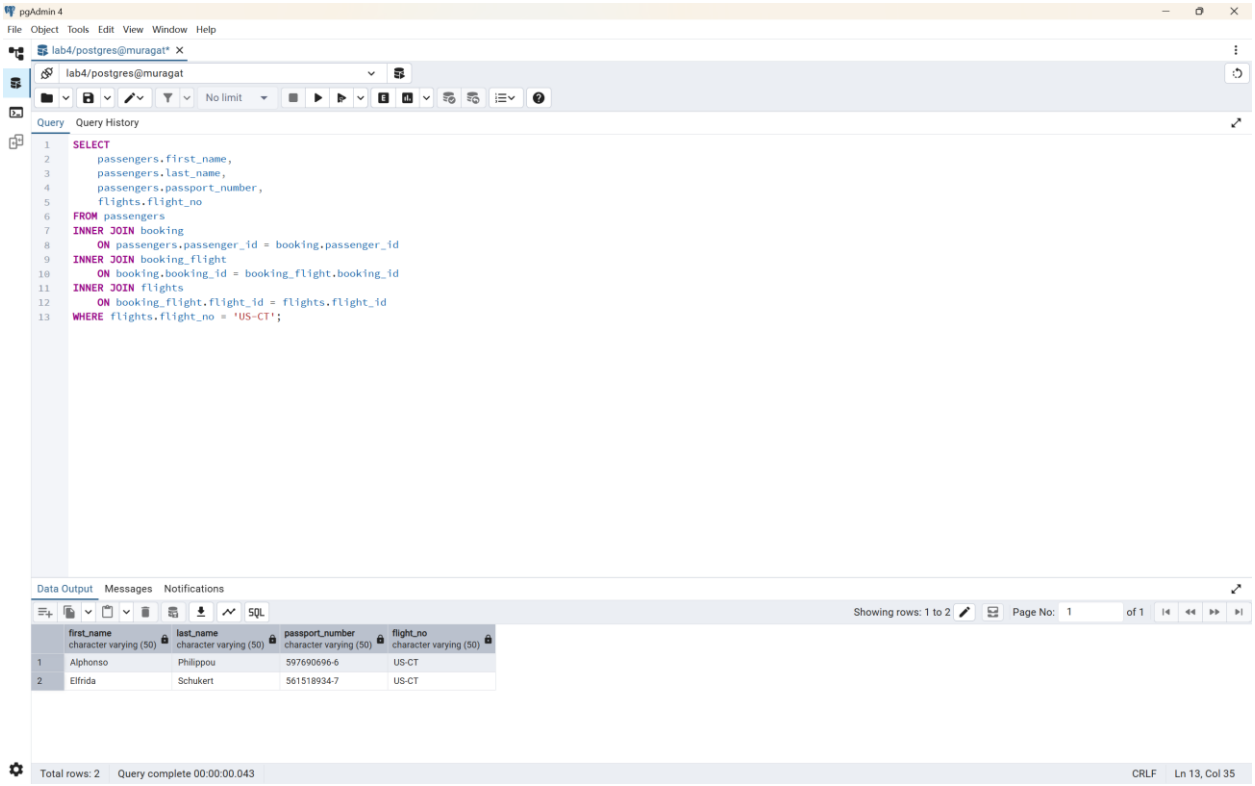
2. Compose a query to obtain a list of all flights with the names of departure airports.



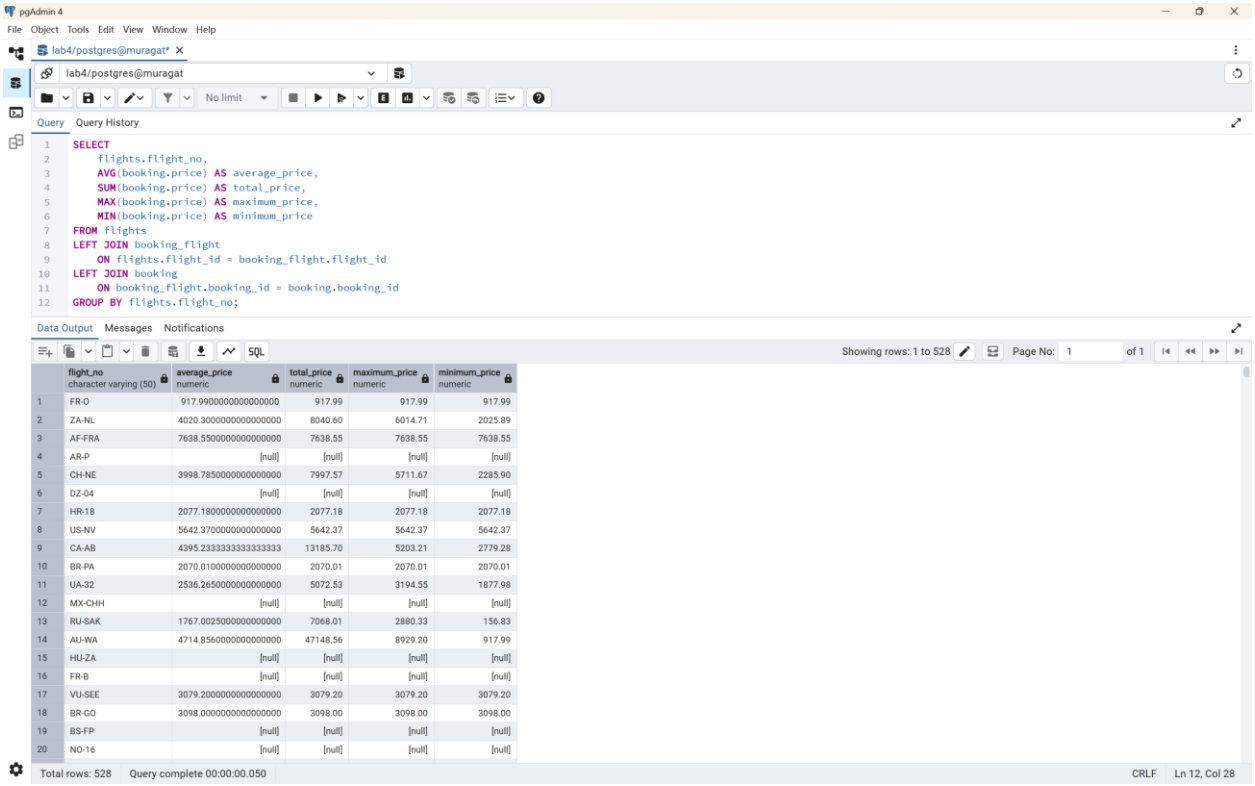
3. Create a query that finds all airlines that have no flights scheduled for the next month.



4. Create a query to display a list of passengers on a specific flight.



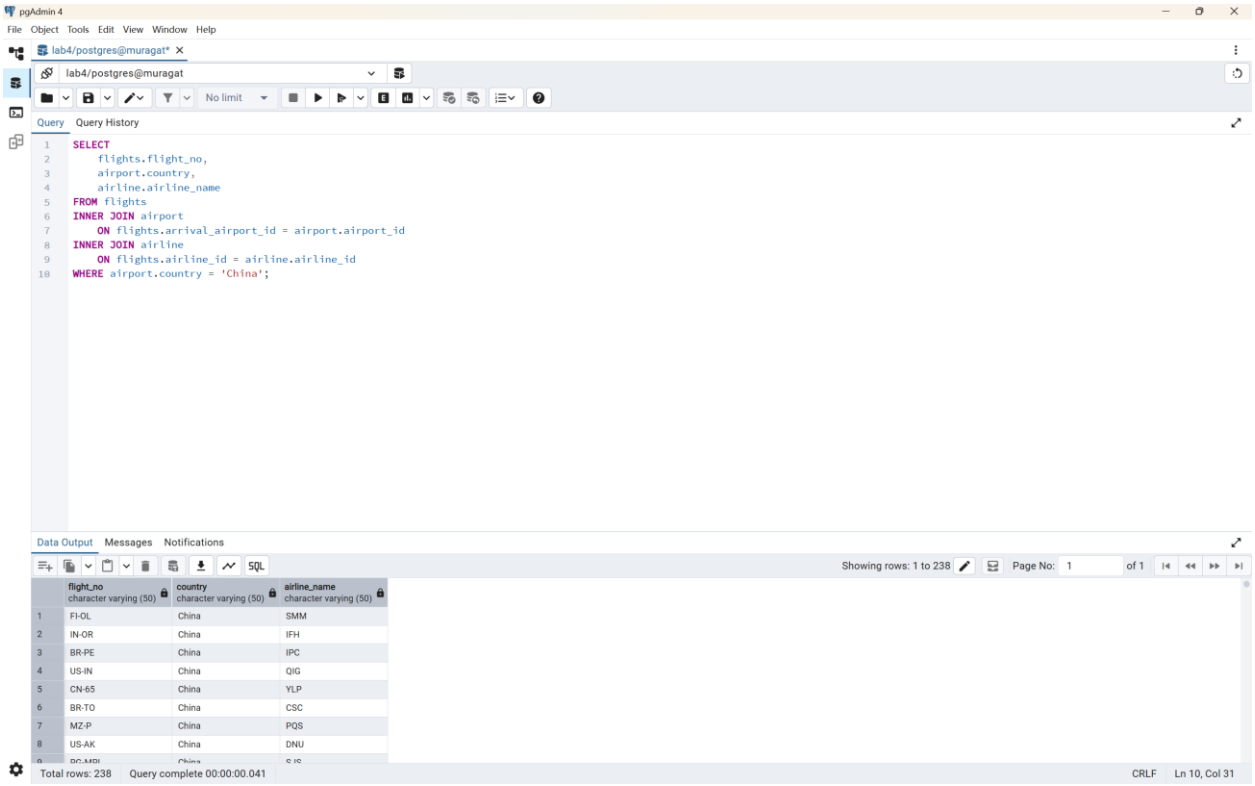
5. Write a query that calculates the average, total, maximum and minimum price of tickets for each flight.



```
SELECT
    flights.flight_no,
    AVG(booking.price) AS average_price,
    SUM(booking.price) AS total_price,
    MAX(booking.price) AS maximum_price,
    MIN(booking.price) AS minimum_price
FROM flights
LEFT JOIN booking_flight
    ON flights.flight_id = booking_flight.flight_id
LEFT JOIN booking
    ON booking_flight.booking_id = booking.booking_id
GROUP BY flights.flight_no;
```

	flight_no character varying (50)	average_price numeric	total_price numeric	maximum_price numeric	minimum_price numeric
1	FR-O	917.9900000000000000	917.99	917.99	917.99
2	ZA-NL	4020.3000000000000000	8040.60	6014.71	2025.89
3	AF-FRA	7638.5500000000000000	7638.55	7638.55	7638.55
4	AR-P	[null]	[null]	[null]	[null]
5	CH-NE	3998.7850000000000000	7997.57	5711.67	2285.90
6	DZ-04	[null]	[null]	[null]	[null]
7	HR-18	2077.1800000000000000	2077.18	2077.18	2077.18
8	US-NV	5642.3700000000000000	5642.37	5642.37	5642.37
9	CA-AB	4395.2333333333333333	13185.70	5203.21	2779.28
10	BR-PA	2070.0100000000000000	2070.01	2070.01	2070.01
11	UA-32	2536.2650000000000000	5072.53	3194.55	1877.98
12	MX-CHH	[null]	[null]	[null]	[null]
13	RU-SAK	1767.0025000000000000	7068.01	2880.33	156.83
14	AU-WA	4714.8560000000000000	47148.56	8929.20	917.99
15	HU-ZA	[null]	[null]	[null]	[null]
16	FR-B	[null]	[null]	[null]	[null]
17	VU-SEE	3079.2000000000000000	3079.20	3079.20	3079.20
18	BR-GO	3098.0000000000000000	3098.00	3098.00	3098.00
19	BS-FP	[null]	[null]	[null]	[null]
20	NO-16	[null]	[null]	[null]	[null]

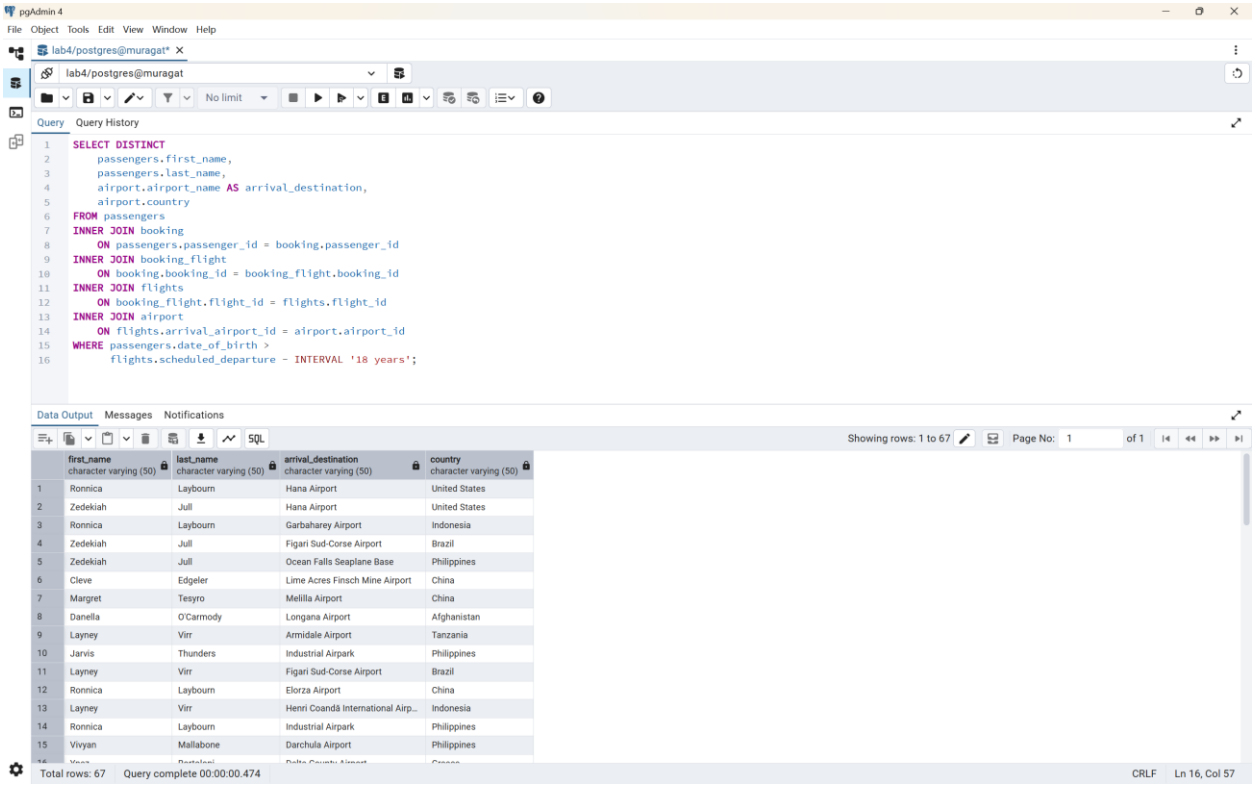
6. Create a query that shows all flights flying to a specific country by combining flights, airports and airline, and using the condition on the country name.



```
SELECT
    flights.flight_no,
    airport.country,
    airline.airline_name
FROM flights
INNER JOIN airport
    ON flights.arrival_airport_id = airport.airport_id
INNER JOIN airline
    ON flights.airline_id = airline.airline_id
WHERE airport.country = 'China';
```

	flight_no character varying (50)	country character varying (50)	airline_name character varying (50)
1	FI-OL	China	SMM
2	IN-OR	China	IFH
3	BR-PE	China	IPC
4	US-IN	China	QIG
5	CN-65	China	YLP
6	BR-TO	China	CSC
7	MZ-P	China	PQS
8	US-AK	China	DNU

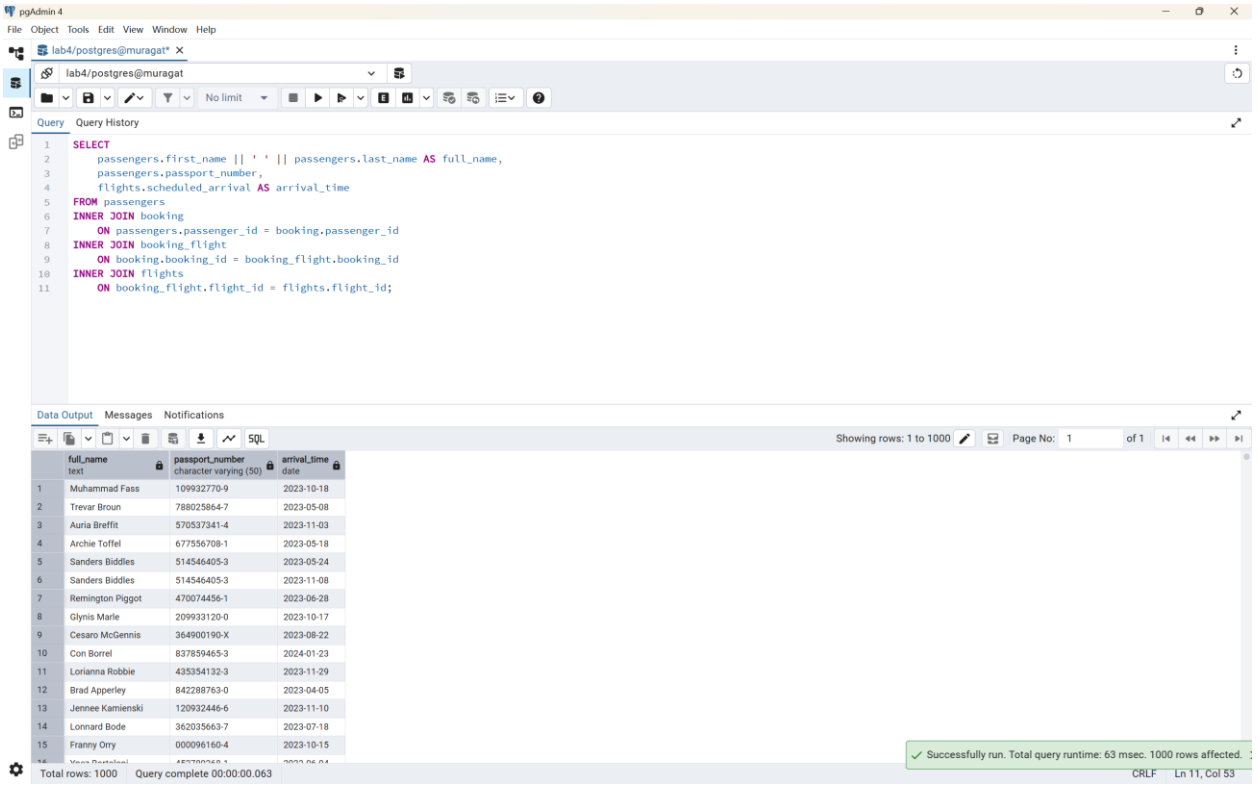
7. Display a list of minor passengers and their arrival destination.



```
SELECT DISTINCT
    passengers.first_name,
    passengers.last_name,
    airport.airport_name AS arrival_destination,
    airport.country
FROM passengers
INNER JOIN booking
    ON passengers.passenger_id = booking.passenger_id
INNER JOIN booking_flight
    ON booking.booking_id = booking_flight.booking_id
INNER JOIN flights
    ON booking_flight.flight_id = flights.flight_id
INNER JOIN airport
    ON flights.arrival_airport_id = airport.airport_id
WHERE passengers.date_of_birth >
      flights.scheduled_departure - INTERVAL '18 years';
```

	first_name character varying (50)	last_name character varying (50)	arrival_destination character varying (50)	country character varying (50)
1	Ronnica	Laybourn	Hana Airport	United States
2	Zedekiah	Jull	Hana Airport	United States
3	Ronnica	Laybourn	Garbaharey Airport	Indonesia
4	Zedekiah	Jull	Figari Sud-Corse Airport	Brazil
5	Zedekiah	Jull	Ocean Falls Seaplane Base	Philippines
6	Cleve	Edgeler	Lime Acres Finsch Mine Airport	China
7	Margret	Tesyro	Melilla Airport	China
8	Danella	O'Carmody	Longana Airport	Afghanistan
9	Layney	Virr	Armidale Airport	Tanzania
10	Jarvis	Thunders	Industrial Airpark	Philippines
11	Layney	Virr	Figari Sud-Corse Airport	Brazil
12	Ronnica	Laybourn	Elorza Airport	China
13	Layney	Virr	Henri Coandă International Airp...	Indonesia
14	Ronnica	Laybourn	Industrial Airpark	Philippines
15	Vivyan	Mallabone	Darchula Airport	Philippines

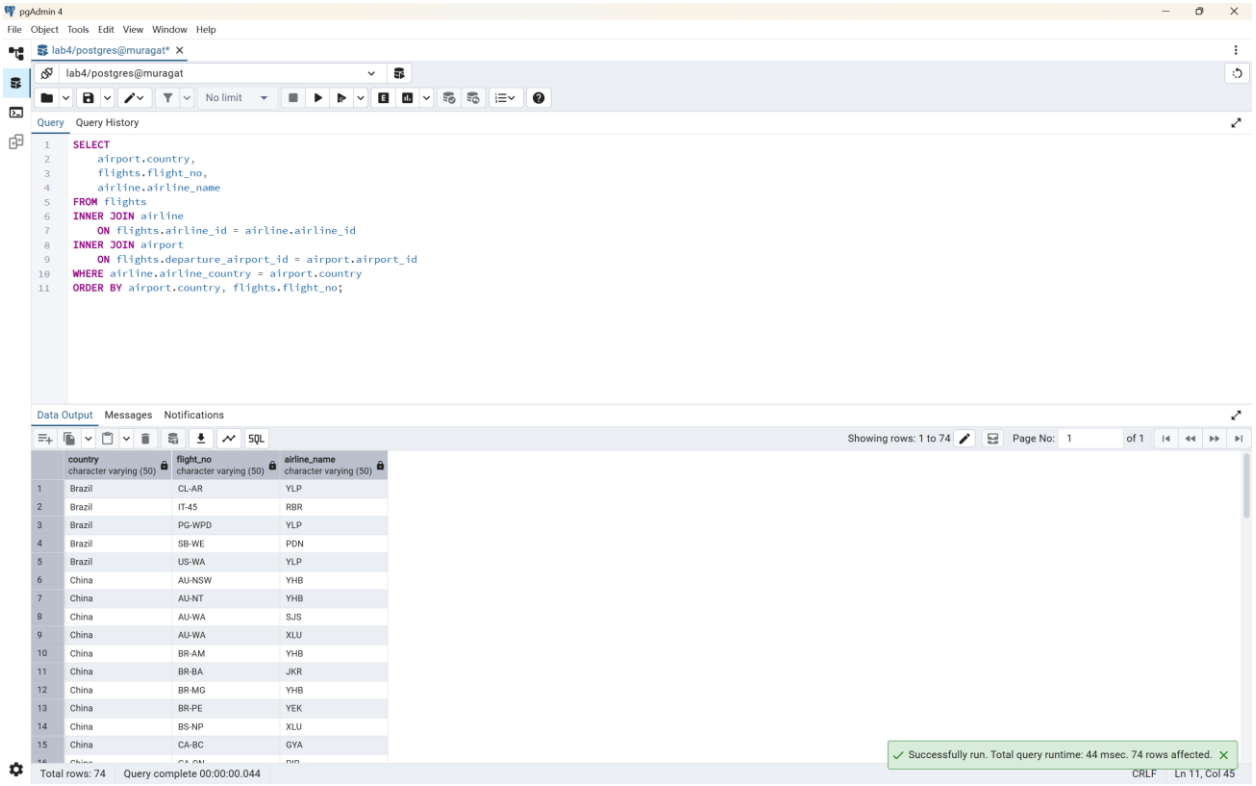
8. Display the passenger's full name, passport number, and the passenger's current time of arrival at the destination.



```
SELECT
    passengers.first_name || ' ' || passengers.last_name AS full_name,
    passengers.passport_number,
    flights.scheduled_arrival AS arrival_time
FROM passengers
INNER JOIN booking
    ON passengers.passenger_id = booking.passenger_id
INNER JOIN booking_flight
    ON booking.booking_id = booking_flight.booking_id
INNER JOIN flights
    ON booking_flight.flight_id = flights.flight_id;
```

	full_name text	passport_number character varying (50)	arrival_time date
1	Muhammad Fass	109932770-9	2023-10-18
2	Trevar Broun	788025864-7	2023-05-08
3	Auria Breffit	570537341-4	2023-11-03
4	Archie Toffel	677556708-1	2023-05-18
5	Sanders Biddles	514546405-3	2023-05-24
6	Sanders Biddles	514546405-3	2023-11-08
7	Remington Piggot	470074456-1	2023-06-28
8	Glynis Marle	209933120-0	2023-10-17
9	Cesaro McGennis	364900190-X	2023-08-22
10	Con Borrel	837859465-3	2024-01-23
11	Lorianna Robbie	435354132-3	2023-11-29
12	Brad Apperley	842288763-0	2023-04-05
13	Jennee Kamienski	120932446-6	2023-11-10
14	Lonnard Bode	362035663-7	2023-07-18
15	Franny Orry	000096160-4	2023-10-15

9. Print a list of flights where the airline's home country and origin country are the same. Group them by the airport country.



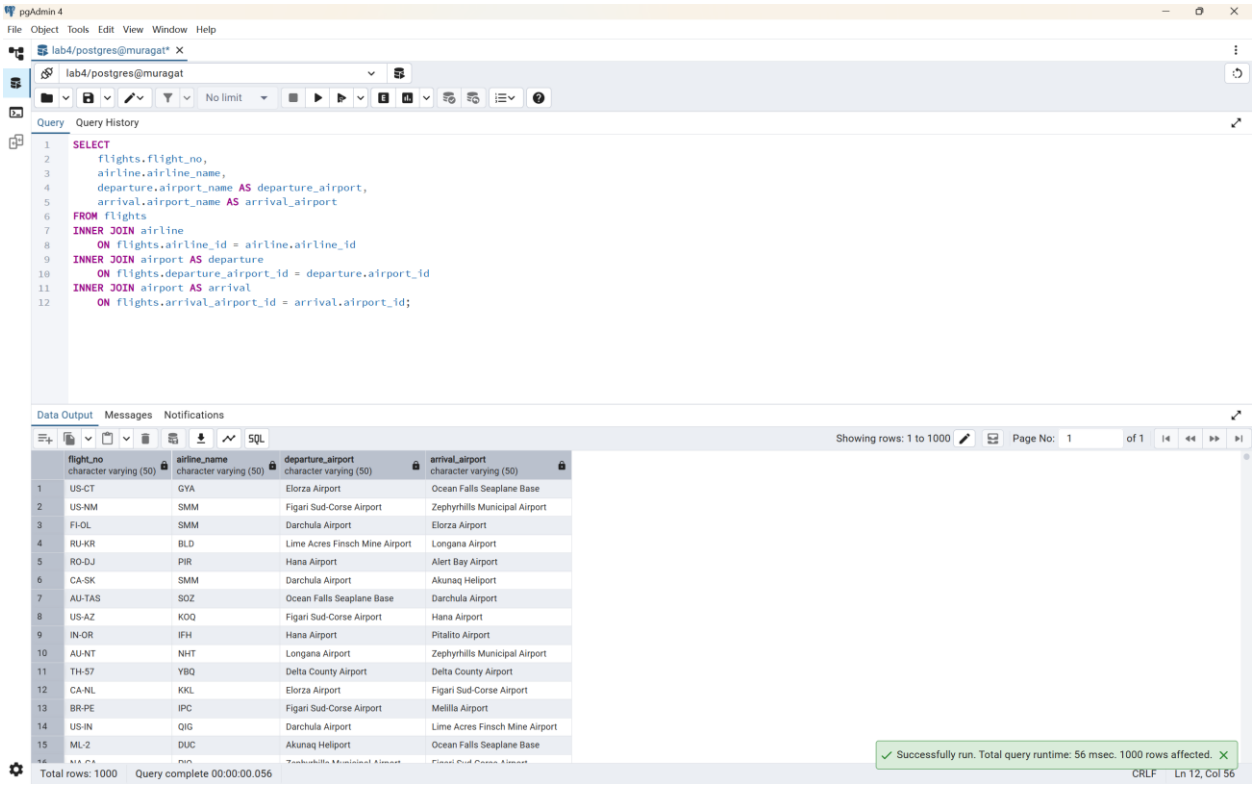
```
SELECT
    airport.country,
    flights.flight_no,
    airline.airline_name
FROM flights
INNER JOIN airline
    ON flights.airline_id = airline.airline_id
INNER JOIN airport
    ON flights.departure_airport_id = airport.airport_id
WHERE airline.airline_country = airport.country
ORDER BY airport.country, flights.flight_no;
```

	country character varying (50)	flight_no character varying (50)	airline_name character varying (50)
1	Brazil	CL-AR	YLP
2	Brazil	IT-45	RBR
3	Brazil	PG-WPD	YLP
4	Brazil	SB-WE	PDN
5	Brazil	US-WA	YLP
6	China	AU-NSW	YHB
7	China	AU-NT	YHB
8	China	AU-WA	SJS
9	China	AU-WA	XLU
10	China	BR-AM	YHB
11	China	BR-BA	JKR
12	China	BR-MG	YHB
13	China	BR-PE	YEK
14	China	BS-NP	XLU
15	China	CA-BC	GYA

Successfully run. Total query runtime: 44 msec. 74 rows affected.

Total rows: 74 Query complete 00:00:00.044 CRLF Ln 11, Col 45

10. Write a query that lists each flight number, airline name, departure airport, and arrival airport by joining the relevant tables.



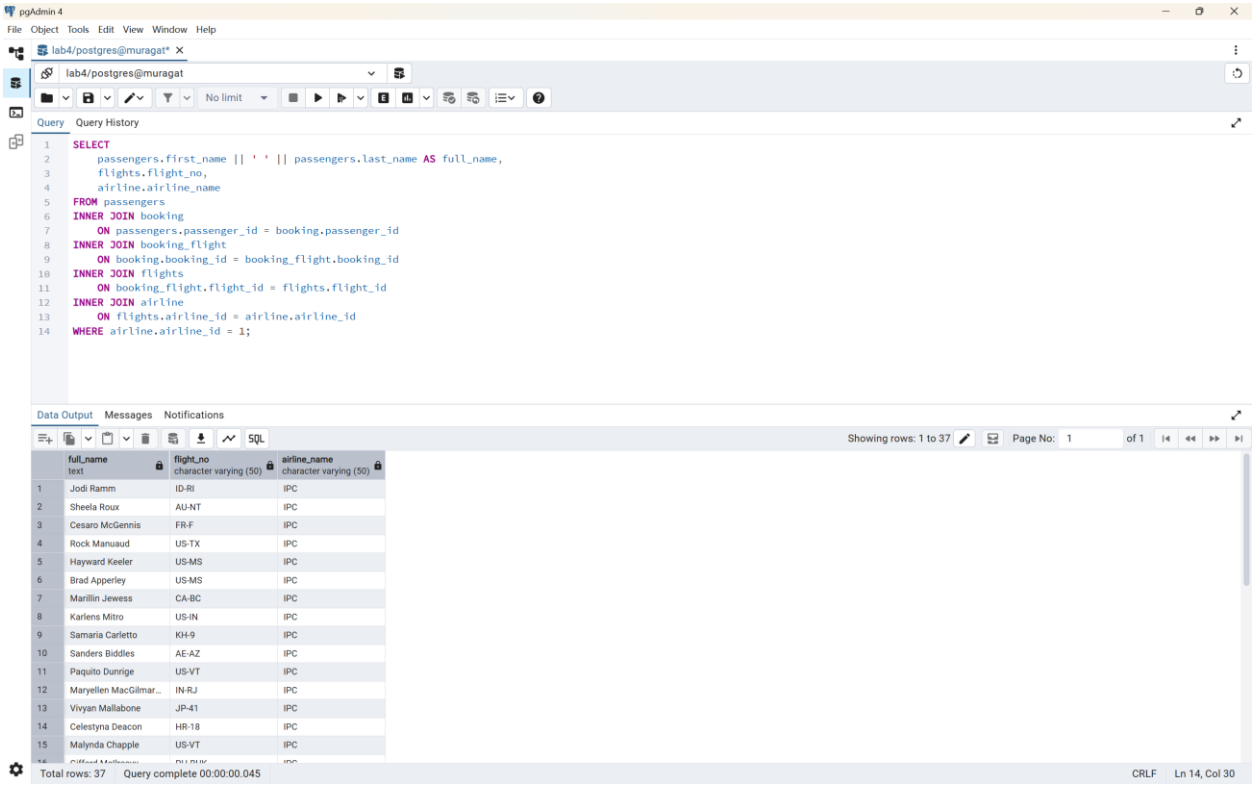
```
SELECT
    flights.flight_no,
    airline.airline_name,
    departure.airport_name AS departure_airport,
    arrival.airport_name AS arrival_airport
FROM flights
INNER JOIN airline
    ON flights.airline_id = airline.airline_id
INNER JOIN airport AS departure
    ON flights.departure_airport_id = departure.airport_id
INNER JOIN airport AS arrival
    ON flights.arrival_airport_id = arrival.airport_id;
```

	flight_no character varying (50)	airline_name character varying (50)	departure_airport character varying (50)	arrival_airport character varying (50)
1	US-CT	GYA	Elorza Airport	Ocean Falls Seaplane Base
2	US-NM	SMM	Figari Sud-Corse Airport	Zephyrhills Municipal Airport
3	FI-OL	SMM	Darchula Airport	Elorza Airport
4	RU-KR	BLD	Lime Acres Finsch Mine Airport	Longana Airport
5	RO-DJ	PIR	Hana Airport	Alert Bay Airport
6	CA-SK	SMM	Darchula Airport	Akunaq Heliport
7	AU-TAS	SOZ	Ocean Falls Seaplane Base	Darchula Airport
8	US-AZ	KOQ	Figari Sud-Corse Airport	Hana Airport
9	IN-OR	IFH	Hana Airport	Pitalito Airport
10	AU-NT	NHT	Longana Airport	Zephyrhills Municipal Airport
11	TH-57	YBQ	Delta County Airport	Delta County Airport
12	CA-NL	KKL	Elorza Airport	Figari Sud-Corse Airport
13	BR-PE	IPC	Figari Sud-Corse Airport	Melilla Airport
14	US-IN	QIG	Darchula Airport	Lime Acres Finsch Mine Airport
15	ML-2	DUC	Akunaq Heliport	Ocean Falls Seaplane Base

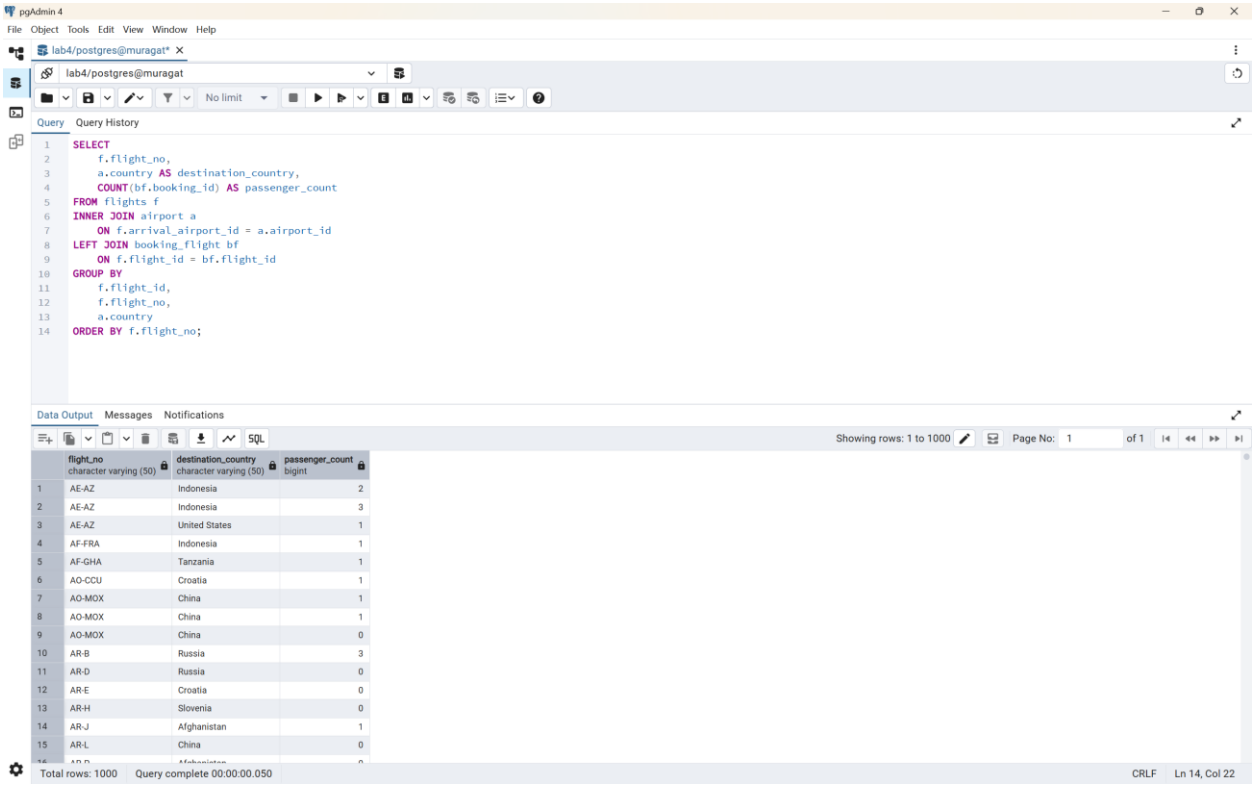
Successfully run. Total query runtime: 56 msec. 1000 rows affected.

Total rows: 1000 Query complete 00:00:00.056 CRLF Ln 12, Col 56

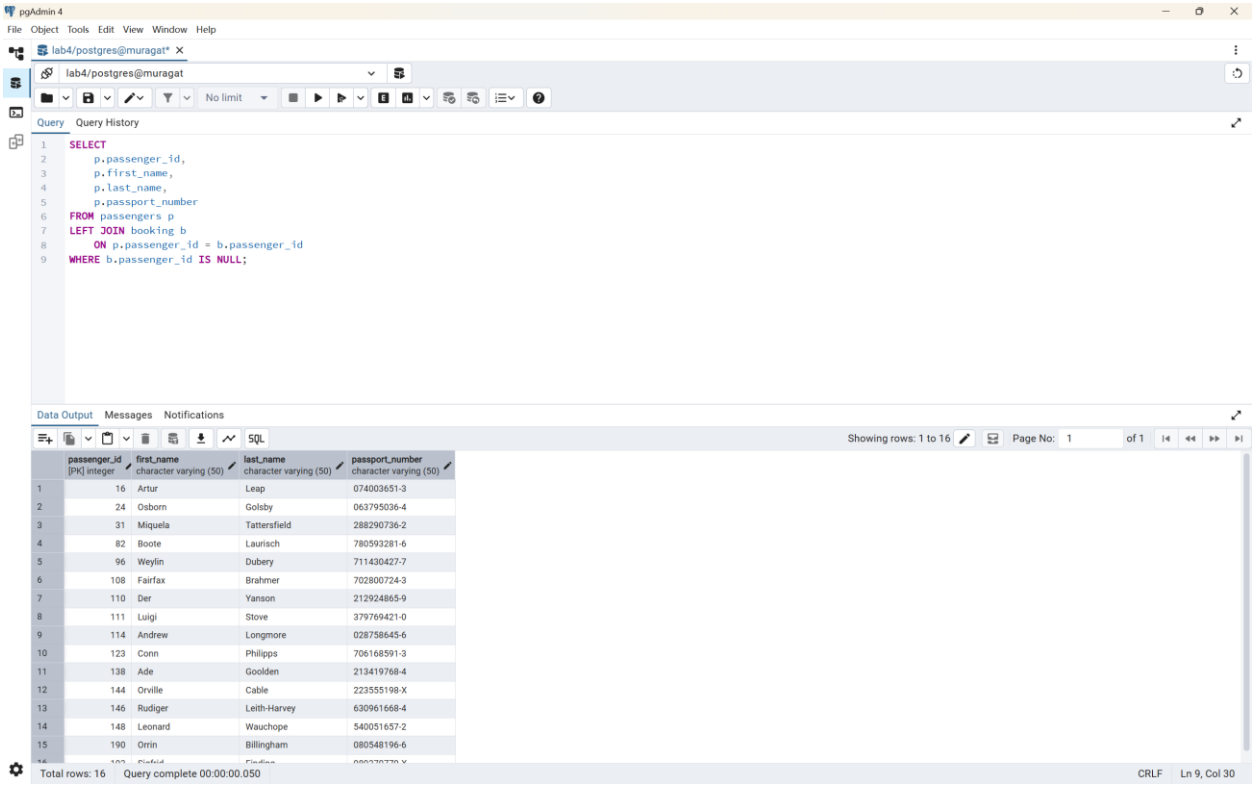
11. Display all passengers who have booked tickets for flights operated by a specific airline, including the passenger's full name and flight number.



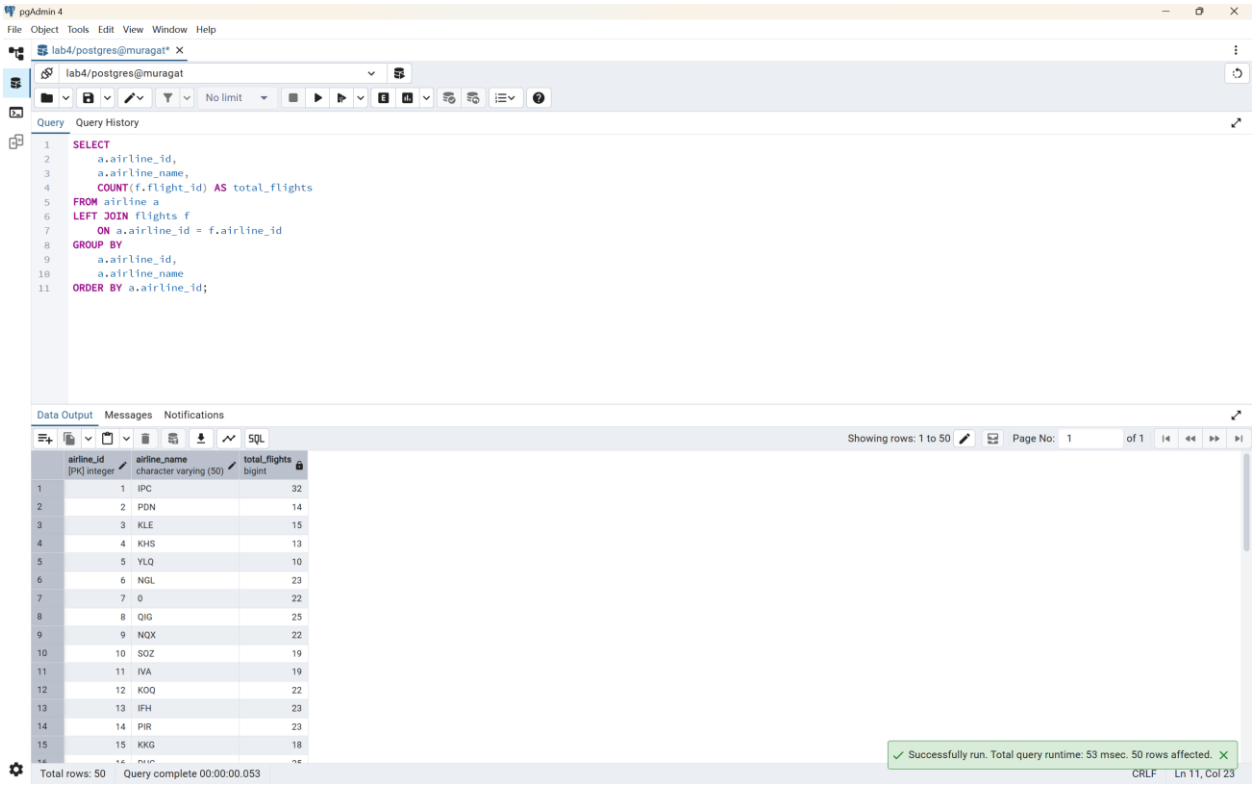
12. Create a query that shows each flight's number, destination country, and number of passengers booked on that flight. Include flights with no booked passengers.



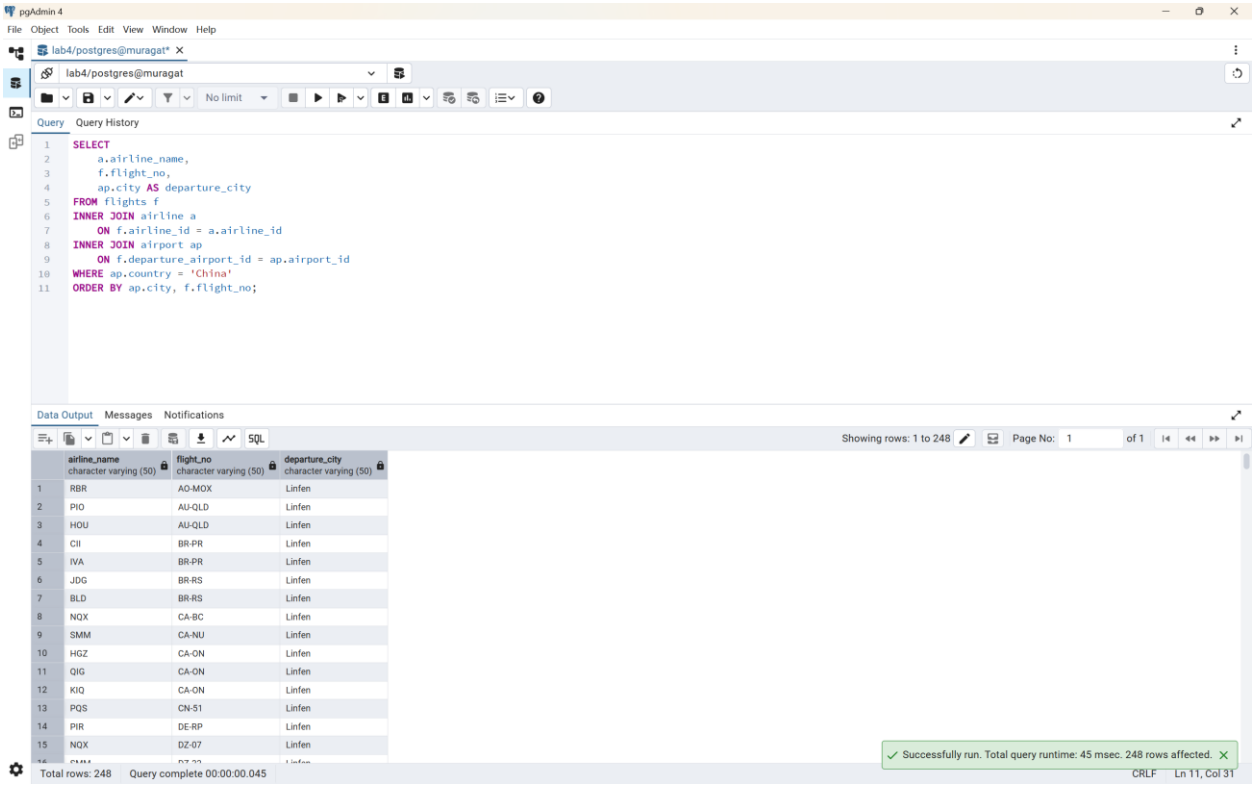
13. Find passengers who have not booked any tickets, using an appropriate JOIN operation.



14. Display each airline and the total number of flights it operates, including airlines that currently have no flights.



15. List flights departing from airports in a specified country and show the airline name, flight number, and departure airport city.


pgAdmin 4
File Object Tools Edit View Window Help
lab4/postgres@muragat*
lab4/postgres@muragat
No limit
Query Query History
SELECT
a.airline_name,
f.flight_no,
ap.city AS departure_city
FROM flights f
INNER JOIN airline a
ON f.airline_id = a.airline_id
INNER JOIN airport ap
ON f.departure_airport_id = ap.airport_id
WHERE ap.country = 'China'
ORDER BY ap.city, f.flight_no;
Data Output Messages Notifications
SQL
Showing rows: 1 to 248
Page No: 1 of 1
airline_name character varying (50)
flight_no character varying (50)
departure_city character varying (50)
1 RBR AO-MOX Linfen
2 PIO AU-QLD Linfen
3 HOU AU-QLD Linfen
4 CII BR-PR Linfen
5 IVA BR-PR Linfen
6 JDG BR-RS Linfen
7 BLD BR-RS Linfen
8 NQX CA-BC Linfen
9 SMM CA-NU Linfen
10 HGZ CA-ON Linfen
11 QIG CA-ON Linfen
12 KIQ CA-ON Linfen
13 PQS CN-51 Linfen
14 PIR DE-RP Linfen
15 NQX DZ-07 Linfen
Successfully run. Total query runtime: 45 msec. 248 rows affected.
Total rows: 248
Query complete 00:00:00.045
CRLF
Ln 11, Col 31